

Thermodynamics — Final Exam

Chula Vista Christian University

Name: _____ Date: _____

Instructions. Answer all seven problems. Show your work for full credit. You may use a calculator. Useful constant: $R = 8.314 \text{ J/(mol}\cdot\text{K)}$.

Part I — Short Answer

1. Answer briefly (1-2 sentences each).

- (a) State the **First Law of Thermodynamics** in words, and write it as an equation for a closed system.
- (b) Define **extensive** and **intensive** properties, and give one example of each.
- (c) What does it mean for a thermodynamic process to be **quasistatic**?
- (d) State the **Second Law of Thermodynamics** in either the Kelvin or the Clausius form.

Part II — Calculations

2. **Ideal gas equation of state.** A sample of 2.00 mol of an ideal gas at $T = 300 \text{ K}$ occupies a volume $V = 0.0500 \text{ m}^3$. Calculate the pressure p .

3. **First law.** A closed system absorbs $Q = 500 \text{ J}$ of heat from its surroundings and does $W = 200 \text{ J}$ of work on the surroundings. Find the change in internal energy ΔU .

4. Carnot efficiency. A Carnot engine operates between a hot reservoir at $T_H = 600$ K and a cold reservoir at $T_C = 300$ K. Determine its efficiency η .

5. Entropy of isothermal expansion. One mole of ideal gas expands isothermally and reversibly at $T = 300 \text{ K}$ from volume V_1 to $V_2 = 2 V_1$. Calculate the change in entropy ΔS of the gas.

6. Otto cycle efficiency. An idealized Otto cycle has compression ratio $r = 8$ and uses a working gas with $\gamma = 1.4$. Compute the thermal efficiency η . (Recall: $\eta = 1 - r^{1-\gamma}$.)

7. Maxwell relation. Starting from the differential of the Helmholtz free energy

$$dF = -S dT - p dV,$$

use the equality of mixed second partial derivatives to derive the Maxwell relation between $(\partial S/\partial V)_T$ and a partial derivative of p .

End of exam — please check that you have answered all seven problems.